

# Supplementary Methods and Simulation Parameters

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## Supplementary Methods and Simulation Parameters

Companion to *An admissibility test for constitutive claims about phenomenal presence: necessary conditions, asymmetric falsification, and a conjunctive return-compatible criterion*.

This document contains the complete generative specification, parameter values, seeds, computational environment, script-to-output mapping, and diagnostic tables required to reproduce every reported result. Section numbers refer to the main article.

### S1. Computational environment

| Item                  | Value                                                     |
|-----------------------|-----------------------------------------------------------|
| Python                | 3.12.3                                                    |
| NumPy                 | 2.4.4                                                     |
| Matplotlib            | 3.10.8 (figure generation only)                           |
| Processor             | single core                                               |
| Total runtime         | approximately 62 minutes across all reported analyses     |
| External dependencies | none beyond those listed; no network access required      |
| Environment file      | <code>requirements.txt</code> , distributed with the code |

### S2. Generative models

All architectures share a common covariate structure. For  $n$  trials, the stimulus category  $X_c$  is drawn uniformly from four levels and the arousal covariate  $X_k$  from a standard normal

distribution. The design matrix is  $Z = [\mathbf{1}, \text{onehot}(X_c), X_k]$ , augmented with the population blocks named in each estimand.

Write  $\text{dr}()$  for an independently drawn covariate driver,

$$\text{dr}() = \Gamma_{X_c} + X_k \gamma^\top, \quad \Gamma \in \mathbb{R}^{4 \times d}, \quad \gamma \in \mathbb{R}^d, \quad \Gamma_{ij}, \gamma_j \sim \mathcal{N}(0, 1),$$

for the noise term

$$\varepsilon(s) = s Z, \quad Z \sim \mathcal{N}(0, I_d) \quad \text{i.i.d. per trial, equivalently } \varepsilon(s) \sim \mathcal{N}(0, s^2 I_d),$$

with  $\varepsilon \equiv \varepsilon(1)$ , and for the coupling matrix

$$W(s) \in \mathbb{R}^{d \times d}, \quad W(s)_{jk} \sim \mathcal{N}(0, s^2) \text{ i.i.d.},$$

**drawn once per simulation replicate and held fixed across all trials of that replicate;** each occurrence of  $W(\cdot)$  in the table below denotes an independent draw. Population dimensionality is  $d = 6$  throughout.

**Category-indexed latent patterns.** Several generators route information through a discrete category. For  $K$  categories write

$$G \in \mathbb{R}^{K \times d}, \quad G_{kj} \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.},$$

sampled **once per replicate and held fixed across trials**, with  $G[c_i]$  denoting the row indexed by the category  $c_i$  of trial  $i$ . Each population and each link draws its own independent  $G$ . The loading matrices  $L \in \mathbb{R}^{q \times d}$  for latent drivers follow the same convention, with  $L_{jk} \sim \mathcal{N}(0, \sigma^2)$  drawn once per replicate and independently for each population. Each architecture draws its own  $\text{dr}()$  for  $A_1$ ,  $B$  and  $A_2$ , denoted  $\delta_{A_1}, \delta_B, \delta_{A_2}$ , and sets  $A_1 = \delta_{A_1} + \varepsilon(1)$  unless stated otherwise.

| Architecture     | Return | Generative equations                                                                                                |
|------------------|--------|---------------------------------------------------------------------------------------------------------------------|
| Recurrent return | Yes    | $B = \delta_B + A_1 W(0.7) + \varepsilon;$<br>$A_2 = \delta_{A_2} + B W(0.7) + \varepsilon$                         |
| Nonlinear return | Yes    | $B =$<br>$\delta_B + \tanh(A_1) W(0.8) + \varepsilon;$<br>$A_2 =$<br>$\delta_{A_2} + \tanh(B) W(0.8) + \varepsilon$ |

| Architecture                    | Return | Generative equations                                                                                                                                                                                                               |
|---------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Latent confounder + weak return | Yes    | $U \sim \mathcal{N}(0, I_3);$<br>$A_1 = \delta_{A_1} + UL + \varepsilon; B = \delta_B + UL' + A_1W(0.15) + \varepsilon;$<br>$A_2 = \delta_{A_2} + UL'' + BW(0.15) + \varepsilon,$<br>with $L$ entries $\sim \mathcal{N}(0, 0.9^2)$ |
| Local persistence               | No     | $B = \delta_B + A_1W(0.7) + \varepsilon;$<br>$A_2 = \delta_{A_2} + A_1W(0.7) + \varepsilon$                                                                                                                                        |
| Feedforward chain               | No     | $B = \delta_B + A_1W(0.7) + \varepsilon;$<br>$C = BW(0.7) + \varepsilon;$<br>$A_2 = \delta_{A_2} + \varepsilon$                                                                                                                    |
| Observed common cause           | No     | $B = \delta_B + \varepsilon; A_2 = \delta_{A_2} + \varepsilon$                                                                                                                                                                     |
| Parallel pathways               | No     | $B = \delta_B + \varepsilon;$<br>$A_2 = 1.3 \Gamma'_{X_c} + X_k \gamma'^\top + \varepsilon$                                                                                                                                        |
| Latent confounder               | No     | $U \sim \mathcal{N}(0, I_3);$ each of $A_1, B, A_2$ receives an independent $UL$ with $L$ entries $\sim \mathcal{N}(0, 0.9^2)$                                                                                                     |
| Latent feedforward state        | No     | $m \sim \mathcal{N}(0, I_5);$ each of $A_1, B, A_2$ receives an independent $mL$ with $L$ entries $\sim \mathcal{N}(0, 0.9^2)$                                                                                                     |
| Shared low-rank factor          | No     | $c \sim \text{Unif}\{0, 1, 2\}; \sigma = G[c]$<br>with $G \in \mathbb{R}^{3 \times d},$<br>$G_{kj} \sim \mathcal{N}(0, 1.6^2); A_1, B$ receive $\sigma W(0.7); A_2$ receives $\sigma$                                              |
| Coarse dependence               | No     | $c = \text{quartile}(A_{1,1});$<br>$B = \delta_B + \varepsilon;$<br>$A_2 = \delta_{A_2} + G[c] + \varepsilon,$<br>$G \in \mathbb{R}^{4 \times d}, G_{kj} \sim \mathcal{N}(0, 1.5^2)$                                               |

| Architecture     | Return | Generative equations                                                                                                                                                                                                                                                                                               |
|------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Token disruption | No     | $c_1 = \text{quintile}(A_{1,.1});$<br>$B = \delta_B + G^{(1)}[c_1] + \varepsilon,$<br>$G^{(1)} \in \mathbb{R}^{5 \times d};$<br>$c_2 = \text{quintile}(B_{.1});$<br>$A_2 = \delta_{A_2} + G^{(2)}[c_2] + \varepsilon,$<br>$G^{(2)} \in \mathbb{R}^{5 \times d},$ both with<br>entries $\sim \mathcal{N}(0, 1.6^2)$ |

Table S1. Generative equations for the twelve architectures of §4.1. Quintile and quartile operators bin the first dimension of the named population at empirical quantiles.

### S3. Four-outcome scenarios and measured effects

The baseline for every scenario is  $A_1 = \delta_{A_1} + \varepsilon$ ,  $B = \delta_B + A_1 W(0.7) + \varepsilon$ ,  $A_2 = \delta_{A_2} + \varepsilon$ , modified as indicated.

| Scenario                                          | Subjects | Trials | Modification             | $\Delta R_{A_1 \rightarrow B}^2$ | $\Delta R_{B \rightarrow A_2}^2$ | $\theta_{\text{chain}}$ |
|---------------------------------------------------|----------|--------|--------------------------|----------------------------------|----------------------------------|-------------------------|
| Absent<br>return,<br>adequate<br>sample           | 20       | 200    | none                     | 0.342                            | −0.005                           | 0.001                   |
| Absent<br>return,<br>reduced<br>sample            | 6        | 200    | none                     | 0.342                            | −0.005                           | 0.001                   |
| Effect<br>within the<br>equiva-<br>lence zone     | 20       | 200    | $A_2 +=$<br>$B W(0.040)$ | 0.342                            | 0.005                            | 0.006                   |
| Effect<br>within the<br>zone,<br>larger<br>sample | 40       | 400    | $A_2 +=$<br>$B W(0.040)$ | 0.342                            | 0.005                            | 0.006                   |

| Scenario                                         | Subjects | Trials | Modification                                                                                                                  | $\Delta R^2_{A_1 \rightarrow B}$ | $\Delta R^2_{B \rightarrow A_2}$ | $\theta_{\text{chain}}$ |
|--------------------------------------------------|----------|--------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|-------------------------|
| Effect near<br>the<br>boundary                   | 20       | 200    | $A_2 +=$<br>$BW(0.085)$                                                                                                       | 0.342                            | 0.017                            | 0.015                   |
| Effect<br>clearly<br>above $\delta_{\min}$       | 20       | 200    | $A_2 +=$<br>$BW(0.300)$                                                                                                       | 0.342                            | 0.094                            | 0.247                   |
| Return<br>present                                | 20       | 200    | $A_2 +=$<br>$BW(0.600)$                                                                                                       | 0.342                            | 0.127                            | 0.577                   |
| High but<br>quantified<br>measure-<br>ment error | 20       | 200    | return at<br>$W(0.600)$ ;<br>then<br>$B +=$<br>$\varepsilon(2.2)$ and<br>$A_2 +=$<br>$\varepsilon(2.2)$                       | —                                | —                                | —                       |
| Unreliable<br>recording<br>of $B$                | 20       | 200    | return at<br>$W(0.600)$ ;<br>then $B \leftarrow$<br>$\varepsilon(6.0)$                                                        | —                                | —                                | —                       |
| Misspecified<br>covariates                       | 20       | 200    | omitted<br>covariate<br>$X_2 \sim$<br>$\mathcal{N}(0, 1)$<br>enters<br>both $B$<br>and $A_2$                                  | —                                | —                                | —                       |
| Latent con-<br>founding                          | 20       | 200    | $U \sim$<br>$\mathcal{N}(0, I_3)$<br>enters $A_1$ ,<br>$B$ and $A_2$<br>with $L$<br>entries $\sim$<br>$\mathcal{N}(0, 0.9^2)$ | 0.664                            | 0.065                            | 0.083                   |

| Scenario                      | Subjects | Trials | Modification                                                     | $\Delta R^2_{A_1 \rightarrow B}$ | $\Delta R^2_{B \rightarrow A_2}$ | $\theta_{\text{chain}}$ |
|-------------------------------|----------|--------|------------------------------------------------------------------|----------------------------------|----------------------------------|-------------------------|
| Failed<br>positive<br>control | 20       | 200    | control<br>channel<br>misaligned<br>by trial<br>permuta-<br>tion | —                                | —                                | —                       |

Table S2. Scenario definitions and measured effects. Measured effects are means of the bias-corrected per-subject estimates over the subjects of one representative study at seed 100000. Entries marked — are scenarios whose generative model is altered by measurement corruption, an omitted covariate, or a broken control channel, for which a single coefficient does not summarize the target effect.

The near-boundary coefficient was selected so that the binding component would sit at its margin: with  $W(0.085)$  the measured effects are 0.017 on the second component against  $\delta_{\min} = 0.01$  and 0.015 on the chain component against  $\delta_{\text{epi}} = 0.02$ . Measured effects are reported for every scenario so that each label can be checked against what the generator produces.

## S4. Fixed parameters

| Parameter                                                   | Value                                                                                                      |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Population dimensionality $d$                               | 6                                                                                                          |
| Stimulus categories                                         | 4, uniform                                                                                                 |
| Arousal covariate                                           | standard normal                                                                                            |
| Noise                                                       | unit-scale Gaussian, independent across trials and dimensions                                              |
| Ridge penalty                                               | 1.0, fixed                                                                                                 |
| Coefficient on $A_1 \rightarrow B$ (four-outcome scenarios) | 0.7                                                                                                        |
| Train/test split                                            | 50/50 within subject, random permutation                                                                   |
| Strata                                                      | stimulus category $\times$ arousal quartile (16 cells); strata with fewer than 3 held-out trials discarded |

| Parameter                                 | Value                                                           |
|-------------------------------------------|-----------------------------------------------------------------|
| Permutations, segment estimands (§4.1)    | 200                                                             |
| Permutations, each episodic link (§4.1)   | 100                                                             |
| Permutations, bias correction (§4.2–§4.4) | 20                                                              |
| Bootstrap resamples over subjects         | 800                                                             |
| Bootstrap interval                        | 90%, percentile                                                 |
| $\delta_{\min}$                           | 0.01 ( $R^2$ -scale components)                                 |
| $\delta_{\text{epi}}$                     | 0.02 (chain component)                                          |
| Reliability gate                          | out-of-sample $R^2(B \sim X + A_1) > 0.10$                      |
| Estimability gate                         | $\geq 5$ training observations per free parameter               |
| Positive-control gate                     | recovery $> 0.5 \delta_{\min}$ on a channel with a known effect |

Table S3. Parameters held fixed across all reported analyses.

## S5. Random seeds

| Analysis                                      | Seed scheme                               |
|-----------------------------------------------|-------------------------------------------|
| Architecture characterization, generative     | $10000 + i$                               |
| Architecture characterization, first segment  | $20000 + i$                               |
| Architecture characterization, second segment | $21000 + i$                               |
| Architecture characterization, episodic       | $30000 + i$                               |
| Four-outcome, subject $s$ of study $i$        | $100000 + 37i + 1000s$                    |
| Four-outcome, effect estimation               | study seed + $50000 + s$                  |
| Four-outcome, positive control                | study seed + $3000 + k$ , $k = 0 \dots 5$ |
| Four-outcome, bootstrap                       | study seed + $7$                          |
| Sample-size sweep, generative                 | $5000 + i$                                |
| Sample-size sweep, tests                      | $6000 + i$ , $6100 + i$ , $6200 + i$      |

Table S4. Seed schemes. Every reported quantity can be re-derived for a single replicate in isolation.

## S6. Sample-size sweep specification

The sweep uses one weak-confounding generator with no return through  $B$ :

$$U \sim \mathcal{N}(0, I_3), \quad L \text{ entries} \sim \mathcal{N}(0, 0.28^2),$$

$$A_1 = \text{dr}() + UL + \varepsilon, \quad B = \text{dr}() + A_1 W(0.7) + UL' + \varepsilon, \quad A_2 = \text{dr}() + UL'' + \varepsilon.$$

Trial counts 250, 500, 1000, 2000, 4000; 200 replicates per size except 4000, which used 80; 100 permutations per segment estimand and 50 per episodic link.

## S7. Diagnostic validation of the positive-control gate

| Quantity                                                               | Value | 95% Wilson interval         |
|------------------------------------------------------------------------|-------|-----------------------------|
| Studies with a genuinely broken control                                | 200   | —                           |
| Correctly routed to not evaluable (sensitivity)                        | 98.0% | [95.0, 99.2]                |
| Missed and routed to an equivalence decision (false-negative rate)     | 2.0%  | [0.7, 5.7]                  |
| Studies with an intact control                                         | 2200  | —                           |
| Routed to not evaluable by the positive-control gate (false positives) | 0     | below simulation resolution |

Table S5. Performance of the positive-control diagnostic. The unreliable-recording scenario is excluded from the intact-control count for the false-positive calculation because it is routed by the reliability gate, which precedes the positive control in the fixed order.

## S8. Script-to-output mapping

| Script                       | Produces                                                                                                                                    | Command                                                                            | Runtime |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------|
| <code>sim_return.py</code>   | shared core module:<br>design matrices,<br>ridge fit,<br>out-of-sample $R^2$ ,<br>strata, Wilson<br>intervals, eleven base<br>architectures | imported, not<br>executed                                                          | —       |
| <code>chain_return.py</code> | segment estimands,<br>conditional nulls,<br>chain-level matching,<br>token-disruption<br>architecture                                       | imported, not<br>executed                                                          | —       |
| <code>run_chain.py</code>    | Table 6, Figure 3                                                                                                                           | <code>python3</code><br><code>run_chain.py 0:12</code><br>200                      | ~24 min |
| <code>make_fig1.py</code>    | Figure 1 (decision<br>flow)                                                                                                                 | <code>python3</code><br><code>make_fig1.py</code>                                  | <1 min  |
| <code>make_fig2.py</code>    | Figure 2<br>(architecture<br>diagram)                                                                                                       | <code>python3</code><br><code>make_fig2.py</code>                                  | <1 min  |
| <code>make_fig3.py</code>    | Figure 3 (component<br>rates, reads<br><code>chain_*.json</code> )                                                                          | <code>python3</code><br><code>make_fig3.py</code>                                  | <1 min  |
| <code>make_fig4.py</code>    | Figure 4 (outcome<br>heat map, reads<br><code>out_*.json</code> )                                                                           | <code>python3</code><br><code>make_fig4.py</code>                                  | <1 min  |
| <code>run_outcomes.py</code> | Table 7, Figure 4                                                                                                                           | <code>python3</code><br><code>run_outcomes.py</code><br><code>0:12 200</code>      | ~13 min |
| <code>run_sweep.py</code>    | Table 8                                                                                                                                     | <code>python3</code><br><code>run_sweep.py</code><br>250,500,1000,2000,4000<br>200 | ~9 min  |

| Script                     | Produces | Command                                                                                           | Runtime     |
|----------------------------|----------|---------------------------------------------------------------------------------------------------|-------------|
| <code>run_thresh.py</code> | Table 9  | <code>python3</code><br><code>run_thresh.py</code><br>0.005 150 and<br>likewise for 0.01,<br>0.02 | ~5 min each |

Table S6. Mapping from script to reported output. Runtimes are for a single core.

## S9. Documented computational deviations

Two cells depart from the nominal replicate counts, for computational budget, and their intervals are reported accordingly:

1. The 4000-trial cell of Table 8 used 80 replicates rather than 200.
2. Table 9 used 150 studies per cell rather than 200.

No other analysis departs from the counts stated in Table S3.

## S10. Component decision rule, including the adverse direction

For each component, a 90% percentile bootstrap interval  $[\ell, u]$  over subjects is compared with the margin  $\delta$ :

| Condition                         | Outcome                          |
|-----------------------------------|----------------------------------|
| $\ell > \delta$                   | survived                         |
| $\ell > -\delta$ and $u < \delta$ | contradicted (negligible effect) |
| $u < -\delta$                     | contradicted (adverse direction) |
| otherwise                         | inconclusive                     |

The third row makes explicit a case the equivalence framing leaves open. An effect reliably below  $-\delta$  is not negligible; it is large in the direction opposite to the one the component asserts, and is therefore incompatible with a required positive predictive gain. It is reported within the contradicted category rather than as a separate outcome.

This case did not arise in any of the 1200 component decisions taken across the twelve scenarios of §4.2, so no reported classification depends on it. It is specified because it can arise in empirical application, where a component may be reliably negative through inverted channel polarity, sign errors in referencing, or genuine suppression.

## S11. Notes on interpretation of the nulls

The residual permutation schemes of §3.5 are exact only under correct specification of the conditional model. Two consequences follow for empirical application.

First, the first-segment null regresses  $A_1$  on  $X$  alone. If  $X$  omits a shared driver of  $A_1$  and  $B$ , the residuals retain that driver and the permutation destroys an association that the null should have preserved, so the test becomes anticonservative. The observed-common-cause and coarse-dependence architectures satisfy the null by construction and confirm nominal behaviour under correct specification; the misspecified-covariate scenario shows the departure.

Second, the second-segment null regresses  $B$  on  $[X, A_1]$ . This is the block that must be preserved: conditioning on stimulus and arousal strata alone would destroy the legitimate  $A_1$ – $B$  association and test a null that no architecture of interest satisfies.

For empirical application a conditional randomization test drawing surrogates from an estimated conditional distribution (Candès et al. 2018) is preferable, since the covariate structure is not known by construction.